Scratch Extensions 04

Recall:

Scratch: *Scratch is a software that uses blocks in a particular order to give instructions. The instructions can be used to create an animation, story, or game. We can use the 'repeat', 'if', and 'forever' blocks to create an animation.*

Animation is a way to add movement to drawings and images of characters and scenes. Scratch is an application that uses code to create animations. We can code sprites to talk, move, and interact with each other.

In this lesson, we will learn to create an interactive animation about the patterns made by stars in the sky.

A cluster of stars in the sky are grouped together to form a pattern which is called a **constellation.** Over centuries, astronomers and scientists have found many star patterns in the sky, all of which have a name. They are often compared to animals or Gods.

We will create a constellation called **Cancer.**

Cancer constellation

Activity 1

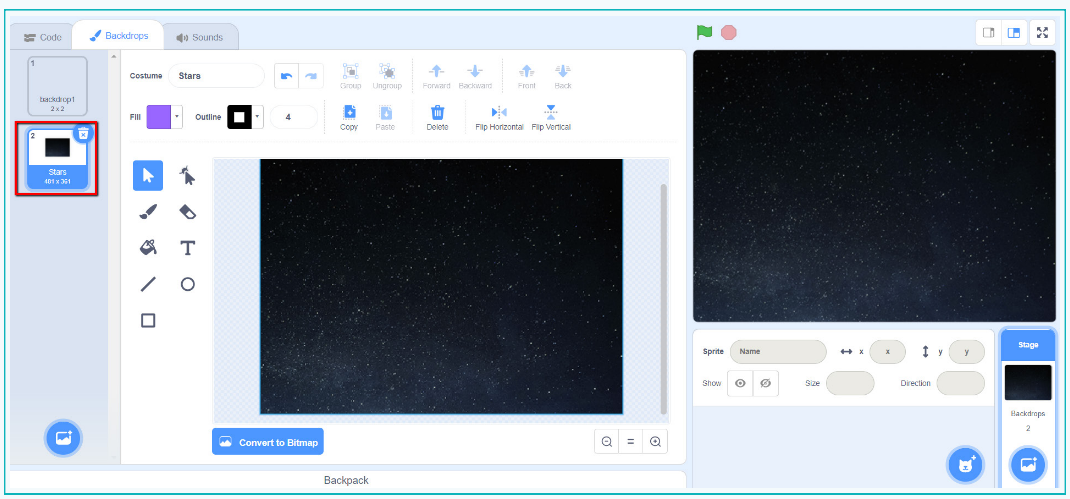
We will create an interactive animation of twinkling stars for the star gazing event.

On a clear, dark night, we can see thousands of stars in the sky. They seem to twinkle, or change their brightness, all the time.

Step 1: Create a backdrop of the night sky for the animation

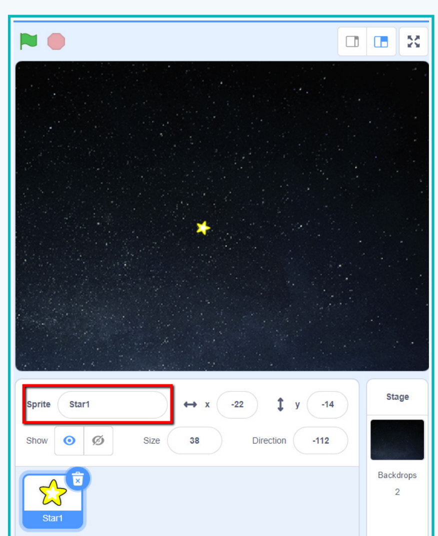
a. Open Scratch.

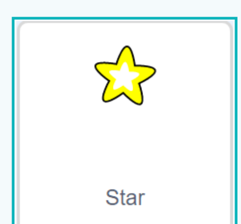
b. Select the 'Stars' backdrop from the library. It should be under the tag 'space'.



c. To create stars, add a 'star' sprite from the library.

d. Name the sprite 'Star1'.





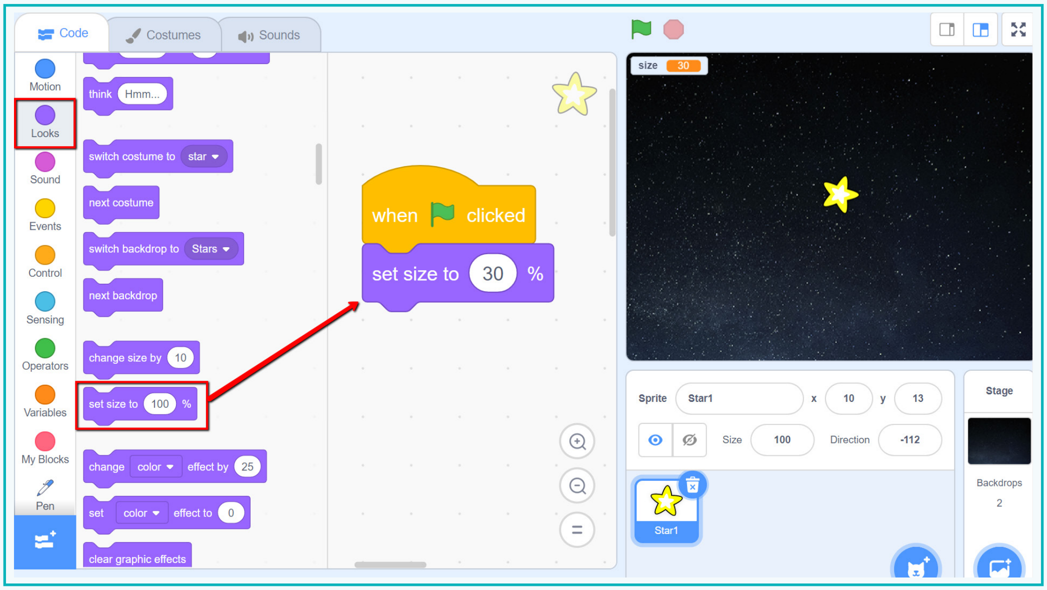
Activity 1

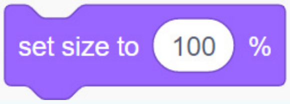
Step 2: Make the star twinkle

To animate the star and make it twinkle, we need to adjust the size of the star. We will increase and then decrease the size rapidly.

a. Add a ‘When 🏳 clicked’ block to start the twinkling movement.

b. Add a ‘Set size to’ block and set the initial size to ‘30%’.





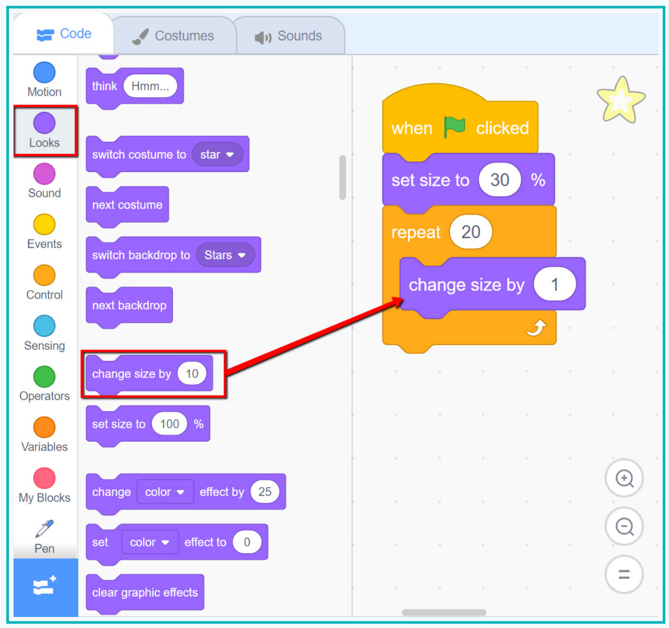
This block sets the size of the sprite. 100% value shows the original size of the sprite.

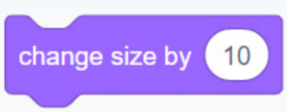
Activity 1

c. To increase the size, add a 'Repeat' block and insert a 'change size by' block.

d. Change the size of the star by '1'.

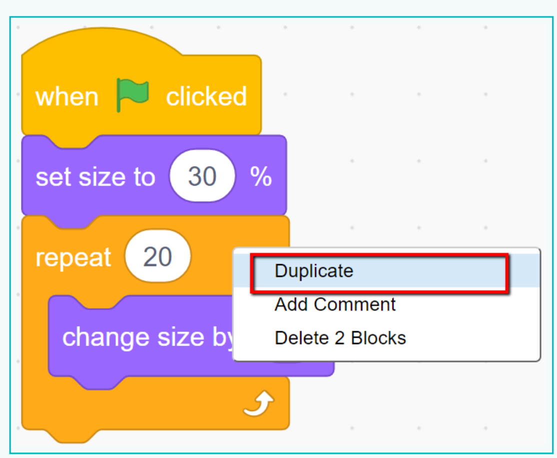
e. Repeat this 20 times.





This block changes its sprite's size by the number specified in the block.

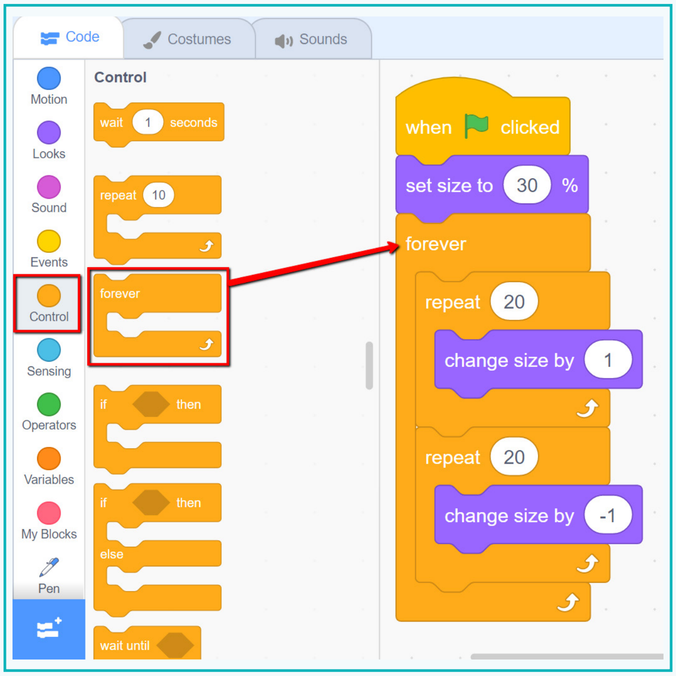
f. To reduce the size, we need a similar block. Right click on the 'repeat' block and duplicate it.



Activity 1

g. In the second 'repeat' block, set the 'change size by' value to '–1'. This will reduce the size.

h. Since stars never stop twinkling, insert both 'repeat' blocks in a 'forever' block.



Recall:

Forever block: *This block repeats the code inside it until the program is completely stopped.*

i. Run the code to watch the star twinkle.

Exercise

Tick mark ☑ the correct answer

1. What will happen when the value in the **change size by** block is –1?

a. The size of the sprite increases by 1 ☐

b. The size of the sprite decreases by 1 ☐

c. The sprite moves in an opposite direction ☐

To make a constellation, we need more than one twinkling star.

Cancer constellation: Cancer is one of the 88 constellations. It represents a crab. It is almost impossible to recognise Cancer as a crab with the naked eye or even with binoculars. It looks more like an upside-down Y. Cancer consists of 5 main stars.

Recall:

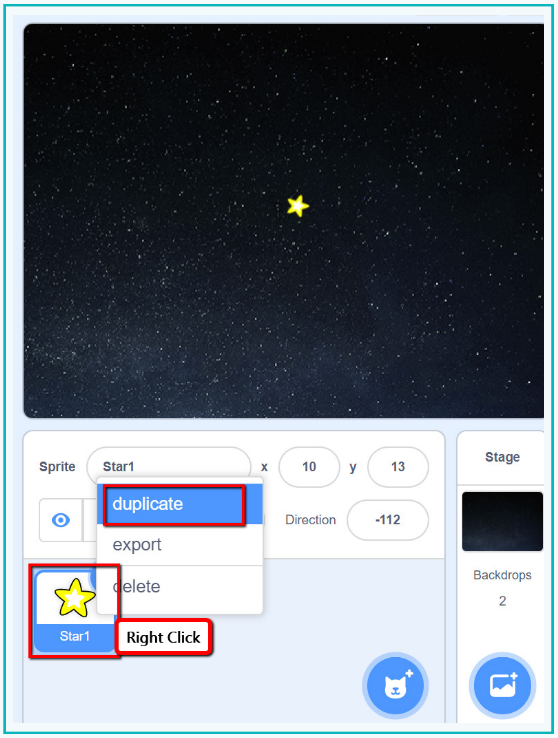
Duplicate: *To duplicate is to make an exact copy of something.*

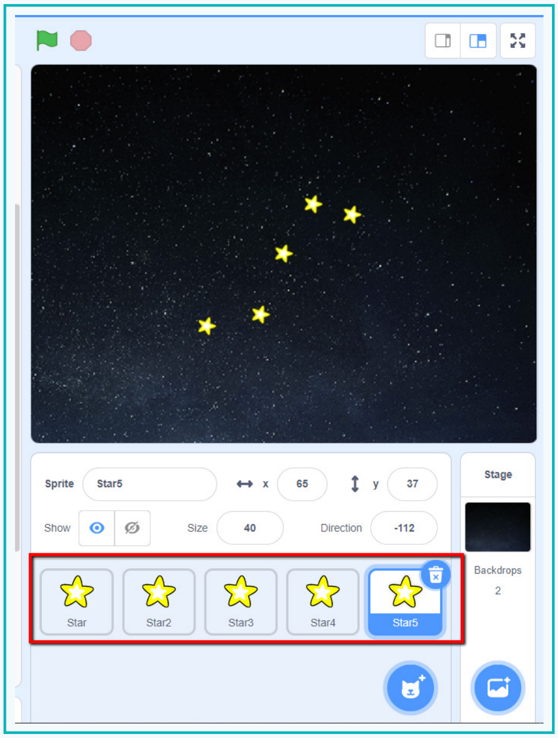
Activity 1

Step 3: Create similar twinkling stars

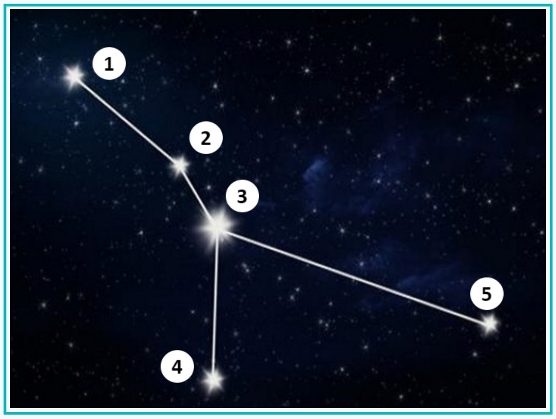
a. Right click on the sprite. From the drop down, select ‘duplicate’.

b. Use ‘duplicate’ to create a total of 5 stars.



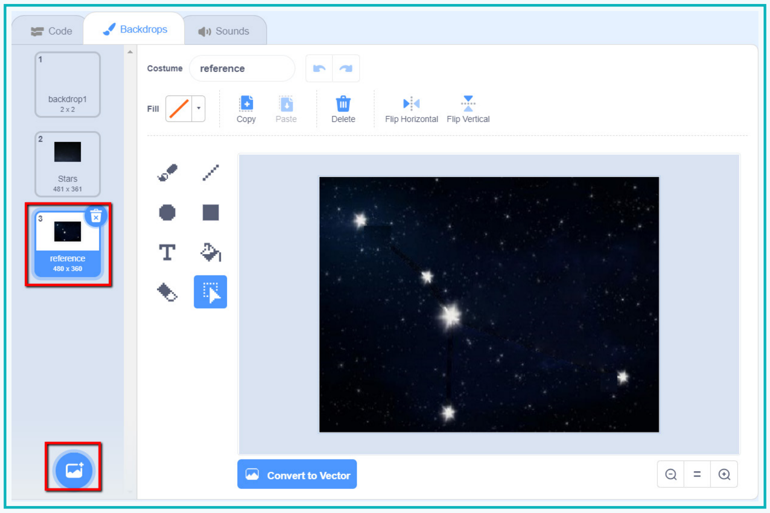


c. Drag and place the stars to resemble the star pattern of the Cancer constellation.



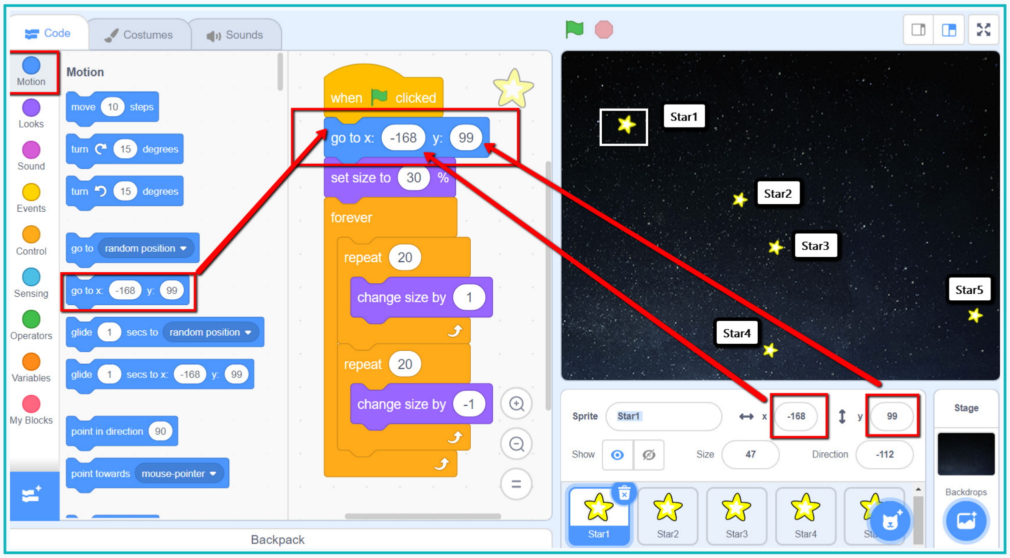
Activity 1

d. An image of the Cancer star pattern can be used as a backdrop for reference.



e. Add a 'go to x y' block at the start for all the star sprites.

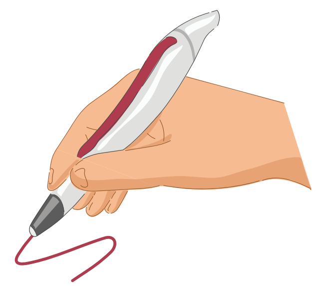
f. Insert values for 'x' and 'y' to arrange the stars in the Cancer pattern.



The stars are now arranged in the Cancer constellation!

To show the complete shape of the constellation, we will use connecting lines to join the stars.

In Scratch, we use **Pen Extension** to add the lines.



Pen Extension: The Pen Extension provides an additional palette to draw, stamp, erase and also to change the colour and size of the pen using code.

When we draw and erase with a pencil on paper, what are the actions we go through?

Step	In real life	In Scratch
01	Sharpen a pencil	Add the pen extension
02	To draw, put the pencil on paper and move it	Use the 'pen down' function
03	To stop drawing, lift up the pencil	Use the 'pen up' function
04	To change the colour, use a different coloured pencil	Use the 'set pen colour' to function
05	To change the thickness, use a pen with thicker nib	Use the 'set pen size' to function
06	To erase, use an eraser to remove everything on the page	Use the 'Erase all' block. Here we cannot erase only one part

The **pen extension** feature can be used to draw anything in Scratch.

Bytes

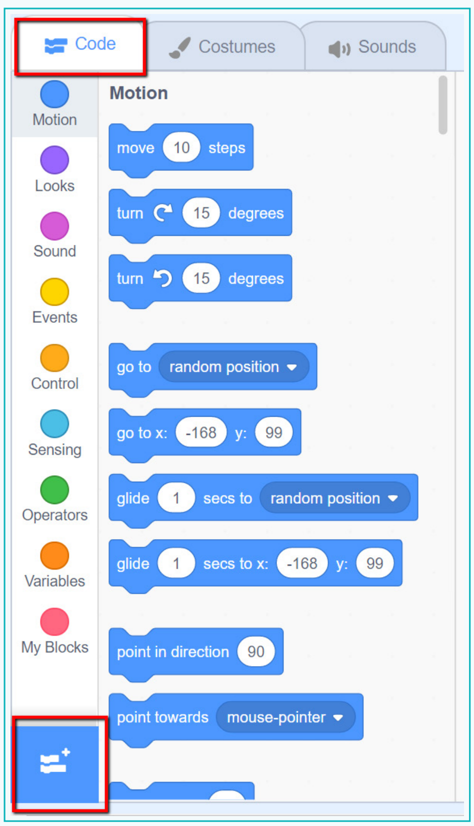
Extension is a part of the software that provides additional functions along with the standard ones.

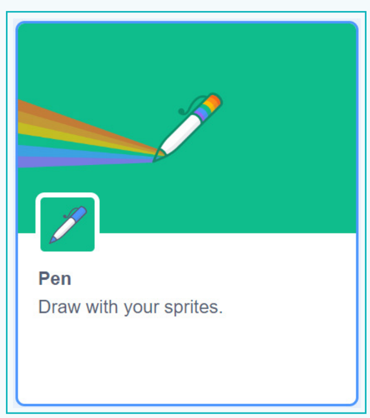
Activity 1

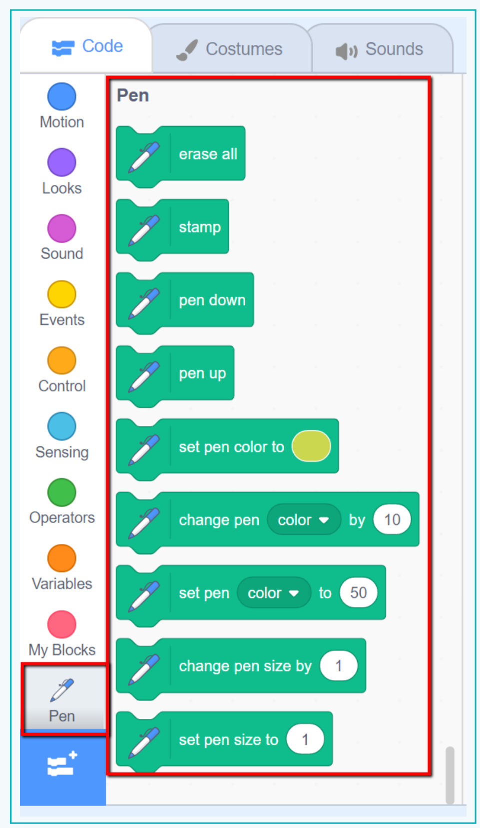
Step 4: Show the exact shape of the star pattern

a. Click on the ‘Extensions’ section in the editor and select the ‘Pen extension’ to enable it.

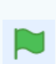
b. An additional palette named ‘Pen’ will be added to the code section.







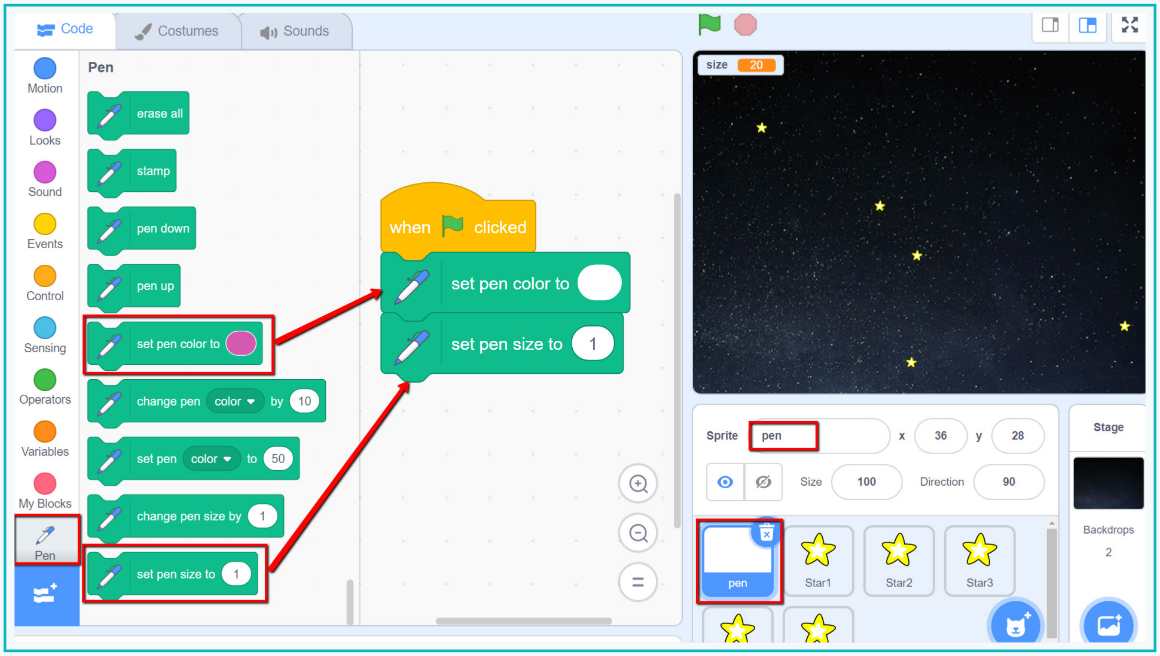
c. To use pen blocks, select ‘paint’ in the ‘choose sprite’ option and create an empty sprite with the name ‘pen’.

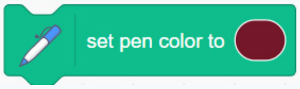
d. Add a ‘When  clicked’ as the starting block.

e. Set the colour of the pen to white, or something light, since our background is black.

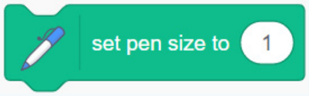
Activity 1

f. Set the ‘pen size’ to ‘1’ for a regular sized line. Increase the size for thicker lines.





This block sets the colour of the pen. To select the colour, click on the coloured area of the block.

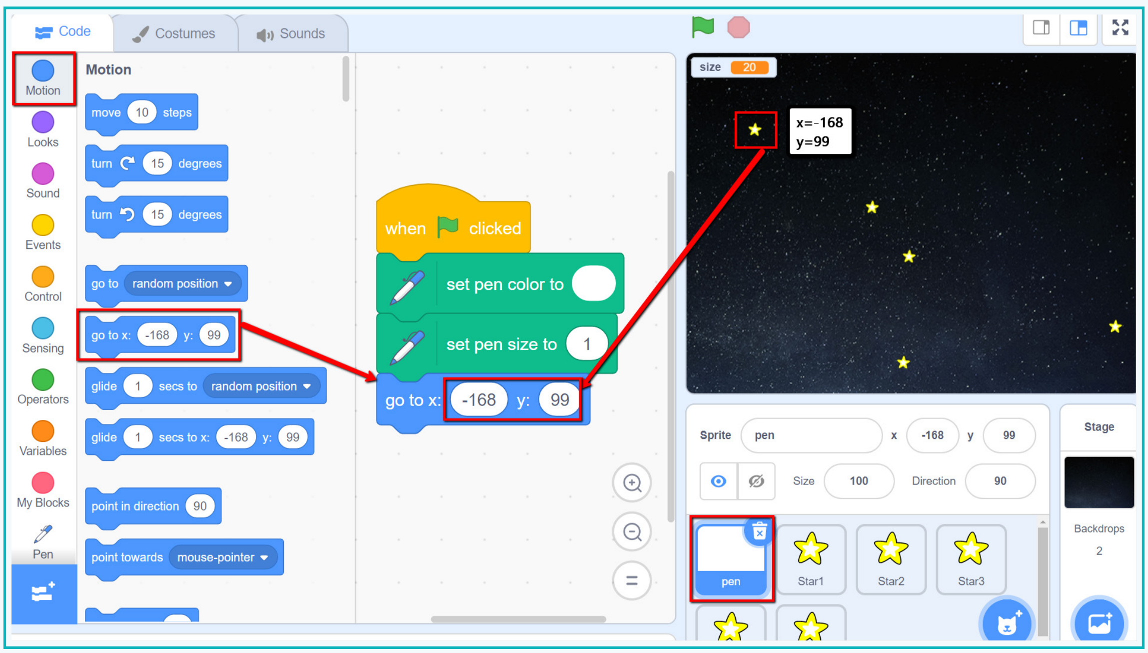


This block sets the size of the pen. The pen’s size determines the thickness of the lines drawn.

g. To start drawing, go to the start point. This will be the position of Star1.

Activity 1

h. Add the ‘go to’ block with the x and y values for ‘Star1’.



This block removes all the marks made by the pen. It is used to clear the screen after the project has started.

This block makes its sprite continuously draw a trail wherever it moves, until the Pen Up block is used.

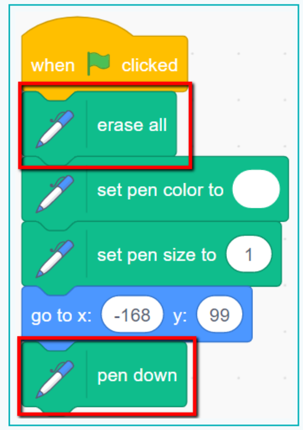
This block makes the sprite stop drawing if the Pen Up block is currently active.

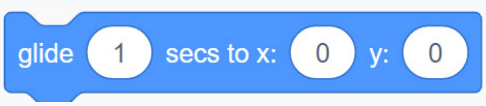
Activity 1

i. To start drawing, add the 'pen down' block from the 'pen palette'. As the sprite moves, it will draw along the path.

j. Add the 'erase all' block at the beginning to erase everything.

k. To draw a line from Star1 to Star2, add the 'glide to' block and enter the x and y values for 'Star2'. Set the 'time' to '2 seconds'.



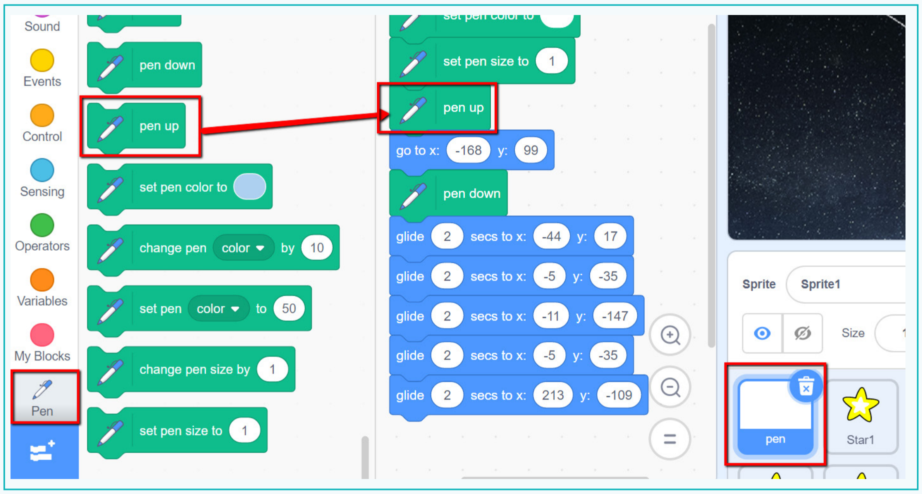


This block moves its sprite to a specified X and Y positions in a specified amount of time.

l. Similarly, add the 'glide to' blocks to follow the path and draw lines to connect the rest of the stars.

Activity 1

m. To avoid unnecessary lines, add the 'pen up' block above the 'go to x y' block.

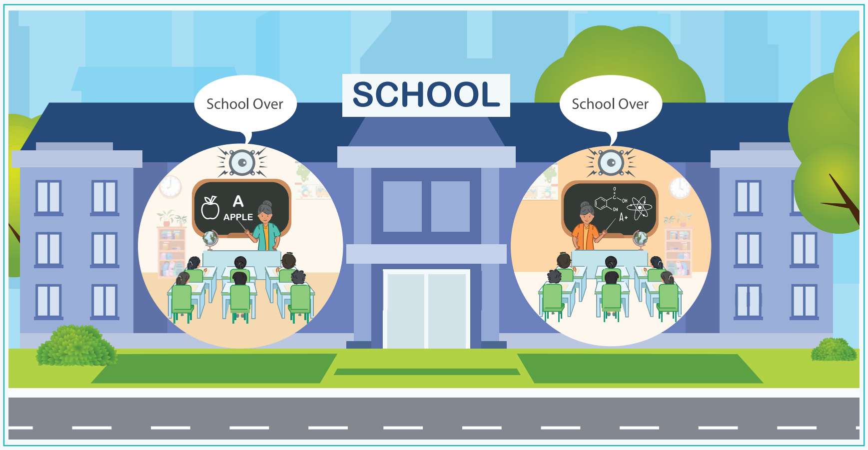


To make an animation interactive, we can ask for user input. Sprites can be used as buttons. If a user clicks on a button sprite, a message is sent to all the other sprites. For this, we use the **broadcast** feature.

Broadcast: A broadcast is a message sent throughout Scratch which is received by other sprites or backdrops as a trigger to run a code script. This feature is useful when we create games or animations where the actions of one sprite trigger the actions of another sprite.

For example:

- In a school, each classroom has a speaker for announcements.
- A message from the principal's office is broadcasted to all the classes. It may contain instructions or announcements for the students.

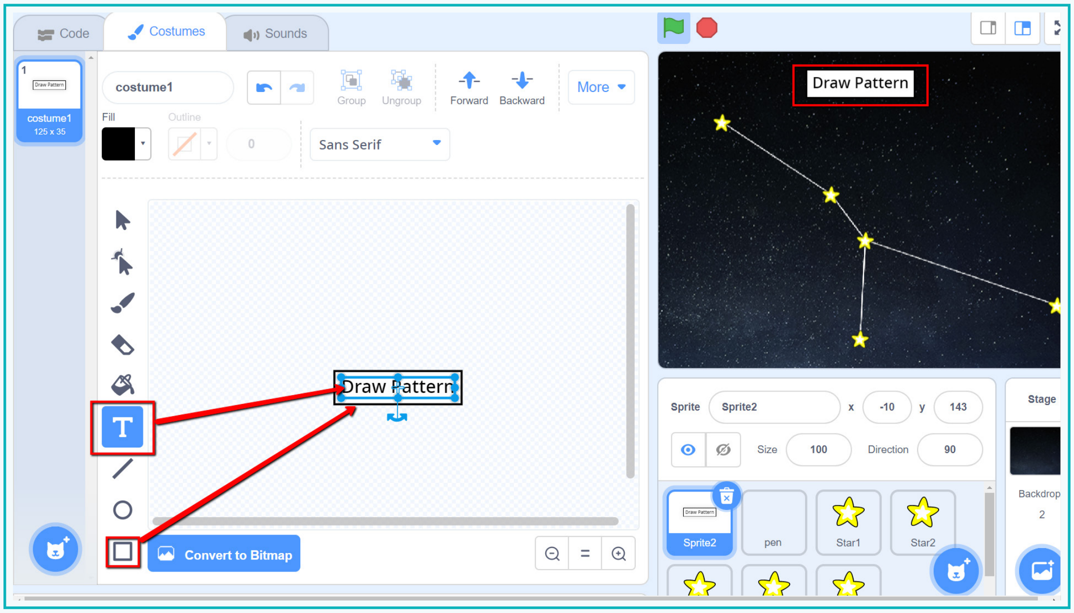


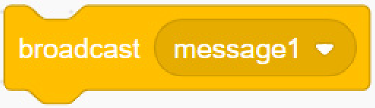
Activity 1

n. We will use the 'broadcast' feature in our constellation animation. If a button sprite is clicked, we will set it to draw a pattern.

o. To create a button sprite, select 'paint' from 'choose a sprite'.

p. Create a button with the text 'draw pattern' and place it at the top of the stage.



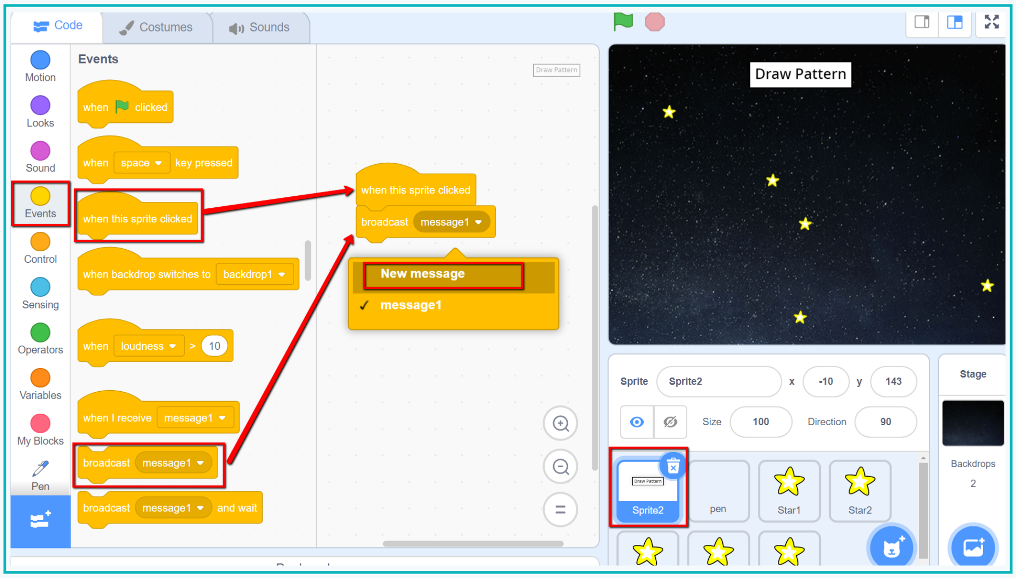


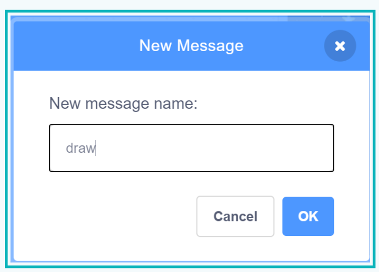
This block broadcasts the specified message throughout the Scratch program. It is used by other sprites to activate a specific code.

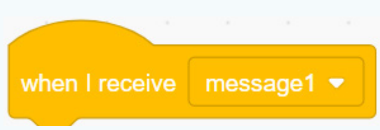
q. Add a 'Broadcast' block. In the dropdown, click on 'new message'.

Activity 1

r. Add a message name. We can call it ‘draw’ or ‘draw pattern’.





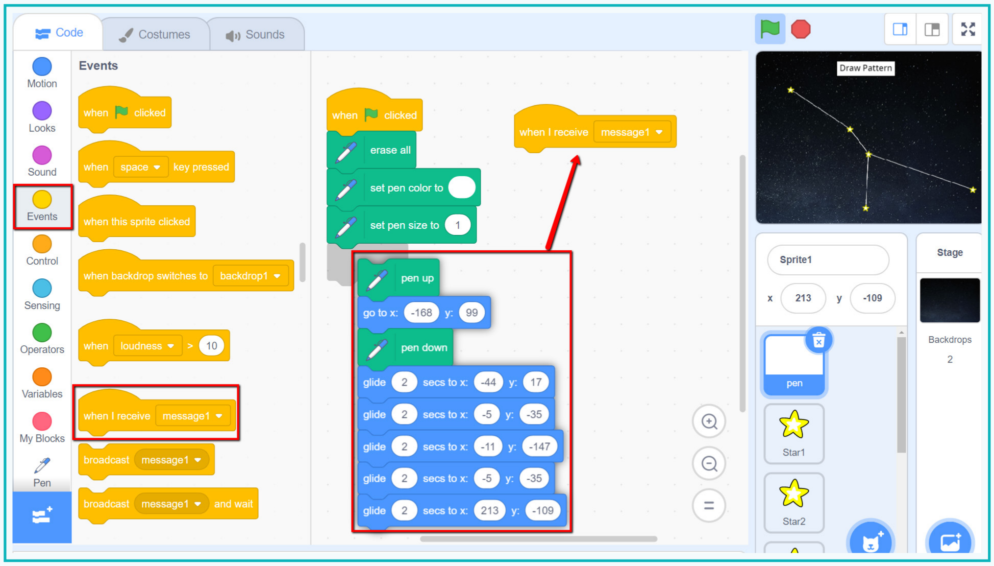


This block is activated when it receives the specified broadcast message. It then runs the script under this block. It stays inactive until the broadcast is received.

s. Add the ‘When I receive’ block to the ‘Pen’ sprite.

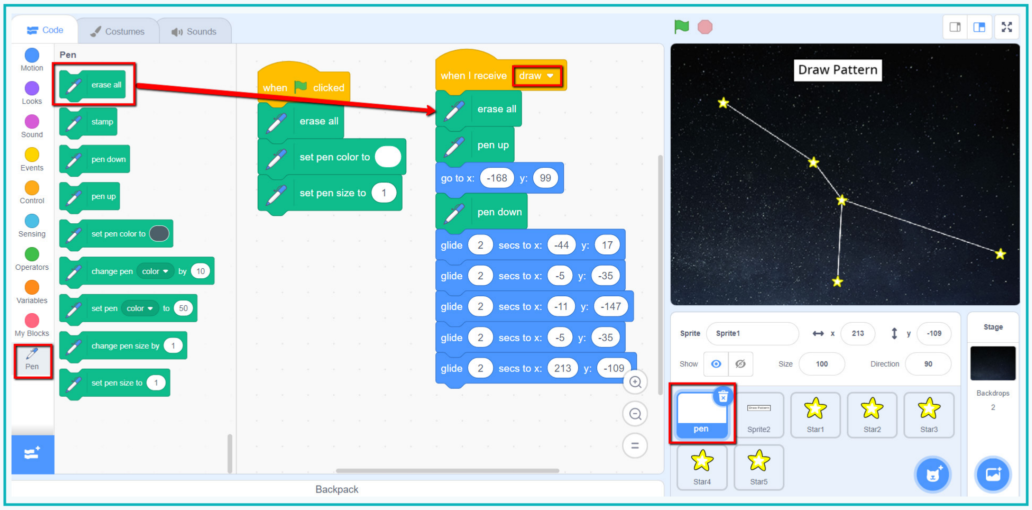
Activity 1

t. Move the code to ‘When I receive’ block to draw the lines.



u. To receive the broadcasted message, select ‘draw’ from the dropdown.

v. Add the ‘erase all’ block to clean the stage before new lines are drawn again.

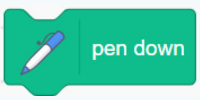


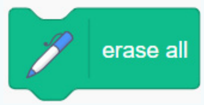
Now, you can create a sprite by yourself to display the name of the constellation ‘Cancer’.

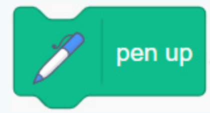
Exercise

Tick mark ☑ the correct answer

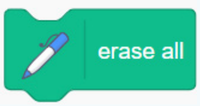
2 Which block is added to stop drawing?

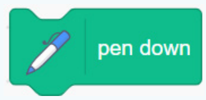
a. ☐ 

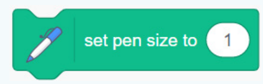
b. ☐ 

c. ☐ 

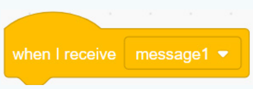
3 Which block is added at the beginning of the code for a clean stage when the code runs?

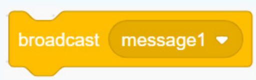
a. ☐ 

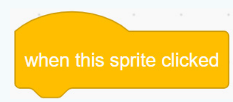
b. ☐ 

c. ☐ 

4 Which block is used to receive a broadcasted message by a sprite or backdrop?

a. ☐ 

b. ☐ 

c. ☐ 

State whether True or False

5 A broadcasted message can be received by any sprite.

▲ ____________________

6 The Pen tool can be set to different sizes and colours.

▲ ____________________

Activity 2

If 🏁 clicked, you can watch as the pattern is drawn. What will happen if we don't add the 'erase all' or 'pen up' blocks at the beginning of the code? Try it out yourself and discuss the outcomes in class.

4. Scratch Extensions

Animation codes sprites to talk, move, and interact with each other. The **Pen** feature in Scratch allows a sprite to draw shapes on the screen with pen blocks. Lines, dots, rectangles, and circles are the easiest shapes to draw. Broadcasts are useful for games and animations since they trigger specific scripts. Scripts are triggered when certain actions are performed. For example, movement of a mouse or a key pressed on the keyboard.

Tech Flash

- **Pen tool** : It uses code to draw, erase, and stamp, similar to the way we draw, colour, erase, and stamp on paper.
- **Broadcast** : This feature sends a message throughout Scratch that can be received by other sprites or backdrops as a trigger to run a code script.